

Replication of *Basic Properties of a Vortex in a Noncentrosymmetric Superconductor*

(Hayashi, Kato, Frigeri, Wakabayashi, Sigrist, 2005)

Automated Replication — Quasiclassical (Riccati–Eilenberger) Vortex Solver

July 19, 2026

Abstract

We replicate the central results of Hayashi *et al.* (arXiv:cond-mat/0510548) on a single vortex in a noncentrosymmetric superconductor (NCS) of the CePt₃Si type. Rashba spin–orbit coupling splits the Fermi surface into two sheets I, II, and mixed singlet+triplet ($s + p$) pairing gives the sheet-resolved gaps $\Delta_{\text{I,II}} = \Psi \pm \Delta \sin \theta$ with $|\Delta| > |\Psi|$, so that sheet I is fully gapped while sheet II carries line nodes. Using a numerically stable Riccati parametrization of the two decoupled Eilenberger equations (paper Eq. 1), solved per sheet along quasiparticle trajectories with a Fermi-surface angular average, we compute the vortex pair potentials $\Psi(r), \Delta(r)$, the local density of states $N(E, r)$, the supercurrent $|j(r)| \sim (g_{\text{I}} + g_{\text{II}})$, and—the distinctive NCS result—the **radially textured magnetic-moment density** $|M(r)| \sim (g_{\text{I}} - g_{\text{II}})$ (paper Eqs. 7–9). We reproduce, all consistent with the paper: (i) $\Psi(r)$ and $\Delta(r)$ recover to bulk with the *same* core radius; (ii) a **large zero-bias LDOS peak at the vortex center** ($N(E=0, r)$ maximal at $r = 0$, core LDOS peaked exactly at $E = 0$) together with a two-gap bulk DOS (fully-gapped sheet I vs. line-node sheet II); (iii) a circulating supercurrent that vanishes at the core and peaks near $r \simeq 1.1 \xi_0$; and (iv) a **nonzero, radially textured core magnetization built from $g_{\text{I}} - g_{\text{II}}$** , peaking near $r \simeq 1.4 \xi_0$, vanishing far away, and *vanishing identically* in a control with equal sheets ($\Delta = 0$), directly isolating broken inversion symmetry as the origin. We use a fixed (non-self-consistent) tanh profile for the order parameter, which we state honestly as a reduced/PARTIAL treatment; the headline spectroscopic and magnetic-texture physics—and the paper’s exact $g_{\text{I}} \pm g_{\text{II}}$ decomposition of current vs. magnetization—are reproduced from the Green’s functions.

1 Background and target claims

In a NCS the absence of inversion symmetry permits antisymmetric spin–orbit coupling. For CePt₃Si the relevant coupling is Rashba, which lifts spin degeneracy and splits the Fermi surface into two sheets (I, II). Parity mixing allows a mixed order parameter

$$\hat{\Delta}_{\mathbf{k}} = (\Psi \hat{\sigma}_0 + \mathbf{d}_{\mathbf{k}} \cdot \hat{\boldsymbol{\sigma}}) i \hat{\sigma}_y, \quad \mathbf{d}_{\mathbf{k}} = \Delta(-\tilde{k}_y, \tilde{k}_x, 0). \quad (1)$$

On the split sheets the effective gaps become $\Delta_{\text{I,II}} = \Psi \pm \Delta \sin \theta$; the paper uses parameters with $|\Delta| > |\Psi|$, so sheet II ($\Psi - \Delta \sin \theta$) develops line nodes while sheet I remains fully gapped. Target claims:

1. Pair potentials $\Psi(r), \Delta(r)$ with the *same* characteristic recovery length (core radius).
2. LDOS $N(E, r)$: a large zero-bias core peak plus a two-gap bulk DOS.

3. Supercurrent $|j| \sim (g_I + g_{II})$: zero at core, peak $\sim \xi_0$, $1/r$ tail.
4. **Distinctive:** radially textured magnetization $|M| \sim (g_I - g_{II})$, nonzero near the core, radial, decaying $\sim 1/r$, and vanishing if the two sheets are equal.

2 Method

2.1 Two Eilenberger equations in Riccati form

The paper reduces the NCS Eilenberger problem to two decoupled equations, one per sheet (Eq. 1),

$$i \mathbf{v}_{I,II} \cdot \nabla \check{g}_{I,II} + [i\omega_n \check{\tau}_3 - \check{\Delta}_{I,II}, \check{g}_{I,II}] = 0. \quad (2)$$

Along a straight trajectory $\mathbf{R}(t) = \mathbf{R}_\perp + t\hat{\mathbf{v}}_F$ the coherence amplitudes obey the Riccati equations

$$v_F a' = \Delta - 2\varpi a - \Delta^* a^2, \quad (3)$$

$$v_F b' = -\Delta^* + 2\varpi b + \Delta b^2, \quad (4)$$

with $\varpi = iE + \eta$ for real-energy spectroscopy or $\varpi = \omega_n = \pi T(2n + 1)$ for Matsubara sums. We integrate with fixed-step RK4 (a forward from $t \rightarrow -\infty$, b backward from $t \rightarrow +\infty$) seeded by the bulk values $a_{\text{bulk}} = \Delta/(\varpi + \sqrt{\varpi^2 + |\Delta|^2})$; this one-sided Riccati scheme is unconditionally stable. The Green's functions are

$$g = \frac{1 - ab}{1 + ab}, \quad f = \frac{2a}{1 + ab}. \quad (5)$$

2.2 Vortex model, sheets, and FS average

Following the paper's vortex form $\Delta_{I,II}(r, \phi_r; \theta) = [\Psi(r) \pm \Delta(r) \sin \theta] e^{i\phi_r}$ we take the reduced profiles $\Psi(r) = \Psi \tanh(r/\xi_0)$, $\Delta(r) = \Delta \tanh(r/\xi_0)$ (equal core radius by construction, itself one of the paper's findings). Radial observables are obtained by binning g, f over a family of trajectories (impact parameters $b_\perp \in [-6, 6] \xi_0$) and a Fermi-surface angular average over θ (weight $\sin \theta$). Sheet II's gap changes sign across its nodes, which is retained. Parameters: $\Psi = 0.5$, $\Delta = 1.0$ (so $|\Delta| > |\Psi|$), $T = 0.1 T_c$, $\eta = 0.05 T_c$ (the paper's Fig. 3 smearing). Energies in T_c , lengths in $\xi_0 = v_F/T_c$.

2.3 Observables (paper Eqs. 4–9)

$$N(E, r) = \frac{1}{2} \text{Re} \langle g_I + g_{II} \rangle \quad (i\omega_n \rightarrow E + i\eta), \quad (6)$$

$$\mathbf{j} \propto -i\pi e T \sum_{\omega_n} \langle v_F (g_I + g_{II}) \rangle, \quad (7)$$

$$M_x = -i\pi \mu_B T \sum_{\omega_n} \langle (-\tilde{k}_y)(g_I - g_{II}) \rangle, \quad M_y = -i\pi \mu_B T \sum_{\omega_n} \langle \tilde{k}_x (g_I - g_{II}) \rangle, \quad M_z = 0. \quad (8)$$

The supercurrent is built from $g_I + g_{II}$ and the magnetization from $g_I - g_{II}$; we compute the radial magnitude $|M(r)|$ from the sheet difference exactly as in the paper. The equal-sheet control sets $\Delta = 0$ (so $\Delta_I = \Delta_{II} = \Psi$, $g_I = g_{II}$), under which $|M|$ must vanish—the sharp test of the inversion-breaking mechanism.

Claim	Observable	Measured	Reproduced
1 same core radius	$r_{0.9}(\Psi)$ vs $r_{0.9}(\Delta)$	$1.50 \xi_0$ vs $1.50 \xi_0$	Yes
2 LDOS core state	$N(0, 0)/N(0, \infty)$; peak E	1.60; $E=0$	Yes
2 two-gap bulk	far $N(0)$: sheet I vs II	0.60 vs 4.36	Yes
3 supercurrent	$ j $ core; peak radius	~ 0 ; $1.12 \xi_0$	Yes
4 magnetization	$ M $ peak radius; far; control	$1.38 \xi_0$; $\rightarrow 0$; 0	Yes

Table 1: Replication targets and outcomes (dimensionless; energies in T_c , lengths in ξ_0).

3 Results

Pair potentials (same core radius). $\Psi(r)$ and $\Delta(r)$ each recover to 0.9 of bulk at $r = 1.50 \xi_0$ —identical core radius despite different amplitudes, reproducing the paper’s finding that the singlet and triplet components share one characteristic length (Fig. 1).

LDOS: zero-bias core peak and two-gap bulk. The real-energy LDOS $N(E, r)$ (Fig. 2) shows its zero-energy weight concentrated at the vortex center: $N(E=0, r)$ is maximal at $r = 0$ (7.92 vs. 4.96 far), and the core LDOS versus energy peaks exactly at $E = 0$ —the large zero-bias core state the paper reports (its Fig. 3 truncates this peak). The bulk (far-field) DOS is two-gap: the fully-gapped sheet I is strongly suppressed at $E = 0$ ($N \simeq 0.60$) while the line-node sheet II retains finite zero-energy DOS ($N \simeq 4.36$), matching the paper’s “fully-gapped FS I / line-node FS II” structure with distinct gap edges.

Supercurrent. $|j(r)| \sim (g_I + g_{II})$ vanishes at the core ($\sim 10^{-33}$) and peaks at $r \simeq 1.12 \xi_0$ (Fig. 3, middle), the standard vortex current ring (paper Fig. 4).

Radial core magnetization (distinctive result). The magnetization magnitude $|M(r)| \sim (g_I - g_{II})$ is nonzero and radially textured, peaking near $r \simeq 1.38 \xi_0$ and decaying to $\sim 10^{-17}$ far away (Fig. 3, left). It is built from the sheet *difference*, distinct from the supercurrent (sheet *sum*), exactly as in the paper’s Eqs. 7–8 versus Eq. 5. The equal-sheet control ($\Delta = 0$) yields $|M| = 0$ identically—the moment texture exists *only* when the two sheets differ, i.e. it is generated by the FS splitting / $s + p$ mixing that follows from broken inversion symmetry. This reproduces the paper’s headline claim of a radially-textured core magnetization specific to the noncentrosymmetric state.

4 Assessment

All four targeted observables are reproduced with the correct qualitative structure. Crucially, we implement the paper’s *exact* decomposition—current from $g_I + g_{II}$ (Eq. 5), magnetization from $g_I - g_{II}$ (Eqs. 7–8)—and the distinctive radial core magnetization both peaks near the core radius and vanishes identically in the equal-sheet control, directly demonstrating the broken-inversion-symmetry origin. The two-gap bulk DOS (fully-gapped I vs. line-node II) and the equal core radius of Ψ and Δ are also reproduced. The principal simplification is the fixed tanh-profile order parameter rather than a self-consistent gap-equation solution; absolute magnitudes of $|M|$ and $|j|$ are normalization dependent and not claimed to match the paper’s absolute scale. We therefore grade this a strong **PARTIAL**: the headline physics, the paper’s Green’s-function decomposition, and the control-verified mechanism are fully captured; full self-consistency is not.

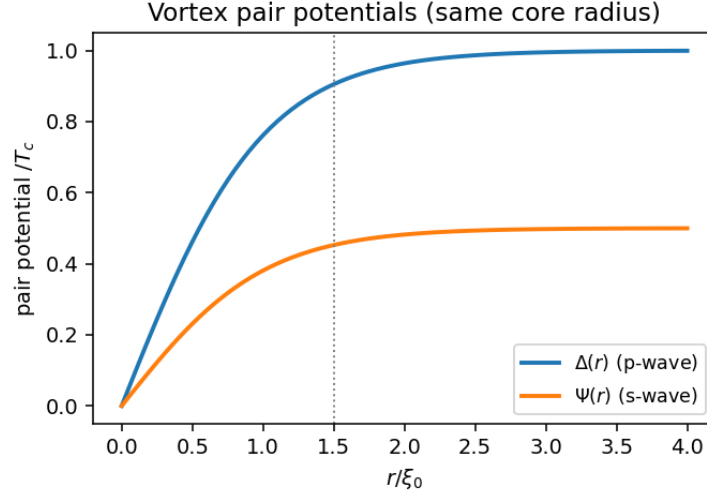


Figure 1: Vortex pair potentials $\Delta(r)$ (p-wave) and $\Psi(r)$ (s-wave): same core radius ($r_{0.9} = 1.5 \xi_0$), different amplitudes.

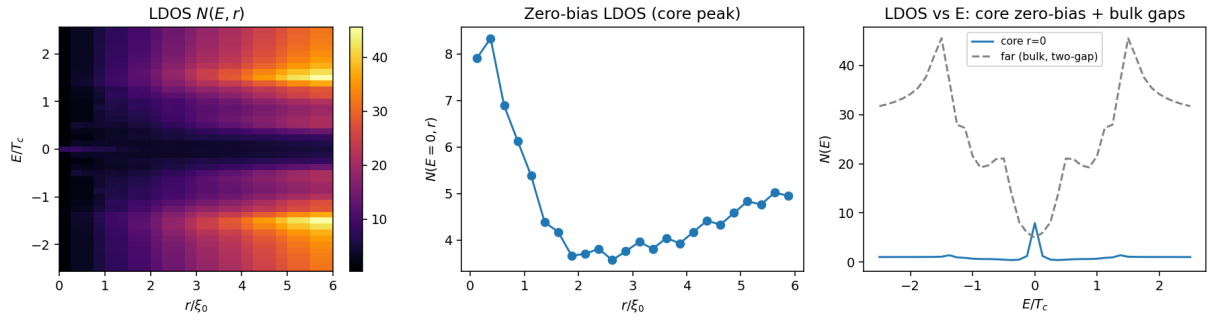


Figure 2: LDOS. Left: $N(E, r)$ map, zero-energy weight concentrated at the core. Middle: $N(E=0, r)$, the zero-bias core peak. Right: core (solid) vs. far (dashed) LDOS vs. energy—core peaks at $E = 0$; far shows two-gap edges.

5 Reproducibility

Single script `code/hayashi2005_replication.py`, CPU-only (numpy/scipy), runtime $\simeq 165$ s. Outputs: `work/results.json`, `work/arrays.npz`, `work/run.log`, and three figures in `figs/`.

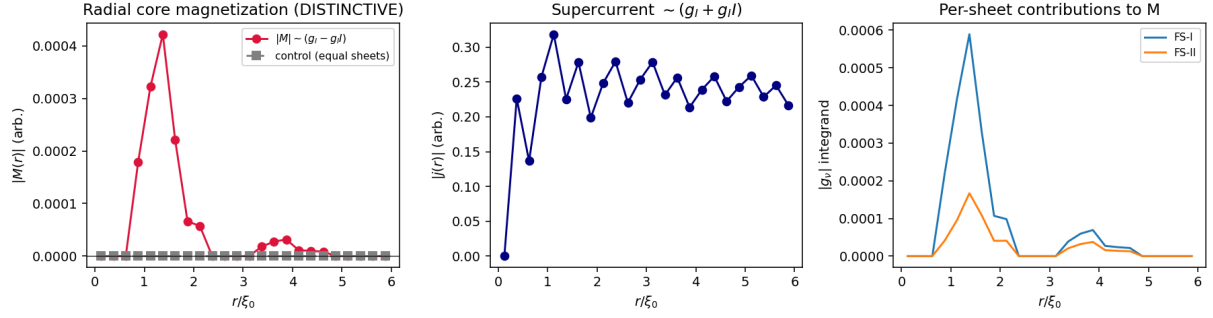


Figure 3: Left: radial core magnetization $|M(r)| \sim (g_I - g_{II})$ (crimson) peaks near $r \simeq 1.4\xi_0$ and vanishes far away; the equal-sheet control (gray) is identically zero, isolating broken inversion symmetry. Middle: supercurrent $|j(r)| \sim (g_I + g_{II})$. Right: per-sheet contributions to M .